Credit Card Clients Delinquency Prediction

Machine Learning Final Project Report

Course: CS 7300 — Machine Learning
Institution: Northeastern University
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Dataset: UCI Default of Credit Card Clients (30,000 records)
Models: Logistic Regression · Hard-Margin SVM · Gaussian Naive Bayes · Neural Network (MLP)

Credit Card Clients Delinquency Prediction

Dataset Link: https://archive.ics.uci.edu/dataset/350/default+of+credit+card+clients

Business Problem Definition:

Financial institutions continuously seek reliable methods to predict client delinquency on credit card payments. Accurate predictions can significantly reduce financial risk, refine lending strategies, and enhance customer service.

This project focuses on developing predictive models from scratch that predict client defaults in the subsequent month based on historical data. By identifying the characteristics and patterns of clients prone to default, preemptive actions can be strategized to diminish potential financial losses.

Problem Setting:

The problem at hand is a binary classification task where the outcome variable (Y) indicates whether a client will default (1) or not (0) in the next month. The challenge lies in utilizing historical payment data, billing statements, and client information to build a robust predictive model without reliance on pre-built libraries for model development.

Data Dictionary:

The dataset is sourced from the UCI Machine Learning Repository and contains 30,000 records and 25 instances. It comprises various features such as credit limit, gender, education, marital status, age, past payment history, bill statements, and previous payment amounts.

This diverse dataset will be leveraged to understand the relationship between client attributes and default behavior. Given the absence of missing values, the dataset offers a suitable foundation for analysis.

Given below is the data dictionary of the dataset: -

Default Payment Next Month (Yes = 1, No = 0)

Sex (1 = male; 2 = female)

Education (1 = graduate school; 2 = university; 3 = high school; 4 = others)

Marriage (1 = married; 2 = single; 3 = others)

Age (years)

Pay_0; . . . ; Pay_6 (-2 = No credit consumed; -1 = pay duly; 1 = payment delay for one month; 2 = payment delay for two months; . . .; 8 = payment delay for eight months)

Bill_Amt1; . . . ;Bill_Amt6 (in NT dollar) (Amount of bill in September 2005; . . . ; Amount of bill in April 2005)

Pay_Amt1; . . .; Pay_Amt6 (in NT dollar) (Amount paid in September 2005; . . . ; Amount paid in April 2005)

	ID	LIMIT_BAL	SEX	EDUCATION	MARRIAGE	AGE	PAY_0	PAY_2	PAY_3	PAY_4	PAY_5	PAY_6	BILL_AMT1	BILL_AMT2	BILL_AMT3	BILL_AMT4	BILL_AMT5	BILL_AMT6	PAY_AMT1	PAY_AMT2	PAY_AMT3	PAY_AMT4	PAY_AMT5	PAY_AMT6	default payment next month
1	1	20000	2	2	1	24	2	2	-1	-1	-2	-2	3913	3102	689	0	0	0	0	689	0	0	0	0	1
2	2	120000	2	2	2	26	-1	2	0	0	0	2	2682	1725	2682	3272	3455	3261	0	1000	1000	1000	0	2000	1
3	3	90000	2	2	2	34	0	0	0	0	0	0	29239	14027	13559	14331	14948	15549	1518	1500	1000	1000	1000	5000	0
4	4	50000	2	2	1	37	0	0	0	0	0	0	46990	48233	49291	28314	28959	29547	2000	2019	1200	1100	1069	1000	0
5	5	50000	1	2	1	57	-1	0	-1	0	0	0	8617	5670	35835	20940	19146	19131	2000	36681	10000	9000	689	679	0
6	6	50000	1	1	2	37	0	0	0	0	0	0	64400	57069	57608	19394	19619	20024	2500	1815	657	1000	1000	800	0
7	7	500000	1	1	2	29	0	0	0	0	0	0	367965	412023	445007	542653	483003	473944	55000	40000	38000	20239	13750	13770	0
8	8	100000	2	2	2	23	0	-1	-1	0	0	-1	11876	380	601	221	-159	567	380	601	0	581	1687	1542	0
9	9	140000	2	3	1	28	0	0	2	0	0	0	11285	14096	12108	12211	11793	3719	3329	0	432	1000	1000	1000	0
10	10	20000	1	3	2	35	-2	-2	-2	-2	-1	-1	0	0	0	0	13007	13912	0	0	0	13007	1122	0	0

Data Cleaning and Pre-Processing

```
0
ID                            0
LIMIT_BAL                     0
SEX                           0
EDUCATION                     0
MARRIAGE                      0
AGE                           0
PAY_0                         0
PAY_2                         0
PAY_3                         0
PAY_4                         0
PAY_5                         0
PAY_6                         0
BILL_AMT1                     0
BILL_AMT2                     0
BILL_AMT3                     0
BILL_AMT4                     0
BILL_AMT5                     0
BILL_AMT6                     0
PAY_AMT1                      0
PAY_AMT2                      0
PAY_AMT3                      0
PAY_AMT4                      0
PAY_AMT5                      0
PAY_AMT6                      0
default payment next month    0
dtype: int64
```

```
<class 'pandas.DataFrame'>
RangeIndex: 30000 entries, 1 to 30000
Data columns (total 25 columns):
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---  ------                      --------------  ----- 
 0   ID                          30000 non-null  object
 1   LIMIT_BAL                   30000 non-null  object
 2   SEX                         30000 non-null  object
 3   EDUCATION                   30000 non-null  object
 4   MARRIAGE                    30000 non-null  object
 5   AGE                         30000 non-null  object
 6   PAY_0                       30000 non-null  object
 7   PAY_2                       30000 non-null  object
 8   PAY_3                       30000 non-null  object
 9   PAY_4                       30000 non-null  object
 10  PAY_5                       30000 non-null  object
 11  PAY_6                       30000 non-null  object
 12  BILL_AMT1                   30000 non-null  object
 13  BILL_AMT2                   30000 non-null  object
 14  BILL_AMT3                   30000 non-null  object
 15  BILL_AMT4                   30000 non-null  object
 16  BILL_AMT5                   30000 non-null  object
 17  BILL_AMT6                   30000 non-null  object
 18  PAY_AMT1                    30000 non-null  object
 19  PAY_AMT2                    30000 non-null  object
 20  PAY_AMT3                    30000 non-null  object
 21  PAY_AMT4                    30000 non-null  object
 22  PAY_AMT5                    30000 non-null  object
 23  PAY_AMT6                    30000 non-null  object
 24  default payment next month  30000 non-null  object
dtypes: object(25)
memory usage: 5.7+ MB
```

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        450000, 60000, 230000, 160000, 280000, 10000, 40000, 210000,
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        30000, 240000, 470000, 480000, 350000, 330000, 110000, 420000,
        170000, 370000, 270000, 220000, 190000, 510000, 460000, 440000,
        410000, 490000, 390000, 580000, 600000, 620000, 610000, 700000,
        670000, 680000, 430000, 550000, 540000, 1000000, 530000, 710000,
        560000, 520000, 750000, 640000, 16000, 570000, 590000, 660000,
        720000, 327680, 740000, 800000, 760000, 690000, 650000, 780000,
        730000], dtype=object),
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 'EDUCATION': array([2, 1, 3, 5, 4, 6, 0], dtype=object),
 'MARRIAGE': array([1, 2, 3, 0], dtype=object),
 'AGE': array([24, 26, 34, 37, 57, 29, 23, 28, 35, 51, 41, 30, 49, 39, 40,
27, 47,
        33, 32, 54, 58, 22, 25, 31, 46, 42, 43, 45, 56, 44, 53, 38, 63,
36,
        52, 48, 55, 60, 50, 75, 61, 73, 59, 21, 67, 66, 62, 70, 72, 64,
65,
        71, 69, 68, 79, 74], dtype=object),
 'PAY_0': array([2, -1, 0, -2, 1, 3, 4, 8, 7, 5, 6], dtype=object),
 'PAY_2': array([2, 0, -1, -2, 3, 5, 7, 4, 1, 6, 8], dtype=object),
 'PAY_3': array([-1, 0, 2, -2, 3, 4, 6, 7, 1, 5, 8], dtype=object),
 'PAY_4': array([-1, 0, -2, 2, 3, 4, 5, 7, 6, 1, 8], dtype=object),
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 'PAY_6': array([-2, 2, 0, -1, 3, 6, 4, 7, 8, 5], dtype=object),
 'BILL_AMT1': array([3913, 2682, 29239, ..., 1683, -1645, 47929],
       shape=(22723,), dtype=object),
 'BILL_AMT2': array([3102, 1725, 14027, ..., 3356, 78379, 48905],
       shape=(22346,), dtype=object),
 'BILL_AMT3': array([689, 2682, 13559, ..., 2758, 76304, 49764],
       shape=(22026,), dtype=object),
 'BILL_AMT4': array([0, 3272, 14331, ..., 20878, 52774, 36535],
       shape=(21548,), dtype=object),
 'BILL_AMT5': array([0, 3455, 14948, ..., 31237, 5190, 32428],
       shape=(21010,), dtype=object),
 'BILL_AMT6': array([0, 3261, 15549, ..., 19357, 48944, 15313],
       shape=(20604,), dtype=object),
 'PAY_AMT1': array([0, 1518, 2000, ..., 10029, 9054, 85900],
       shape=(7943,), dtype=object),
 'PAY_AMT2': array([689, 1000, 1500, ..., 2977, 111784, 3526],
       shape=(7899,), dtype=object),
 'PAY_AMT3': array([0, 1000, 1200, ..., 349395, 8907, 25128],
       shape=(7518,), dtype=object),
 'PAY_AMT4': array([0, 1000, 1100, ..., 2556, 10115, 8049], shape=(6937,),
dtype=object),
 'PAY_AMT5': array([0, 1000, 1069, ..., 8040, 3319, 52964], shape=(6897,),
dtype=object),
 'PAY_AMT6': array([0, 2000, 5000, ..., 70052, 220076, 16080],
       shape=(6939,), dtype=object),
 'default payment next month': array([1, 0], dtype=object)}
```

	LIMIT_BAL	SEX	EDUCATION	MARRIAGE	AGE	PAY_0	PAY_2	PAY_3	PAY_4	PAY_5	PAY_6	BILL_AMT1	BILL_AMT2	BILL_AMT3	BILL_AMT4	BILL_AMT5	BILL_AMT6	PAY_AMT1	PAY_AMT2	PAY_AMT3	PAY_AMT4	PAY_AMT5	PAY_AMT6	default payment next month
1	20000	2	2	1	24	2	2	-1	-1	-2	-2	3913	3102	689	0	0	0	0	689	0	0	0	0	1
2	120000	2	2	2	26	-1	2	0	0	0	2	2682	1725	2682	3272	3455	3261	0	1000	1000	1000	0	2000	1
3	90000	2	2	2	34	0	0	0	0	0	0	29239	14027	13559	14331	14948	15549	1518	1500	1000	1000	1000	5000	0
4	50000	2	2	1	37	0	0	0	0	0	0	46990	48233	49291	28314	28959	29547	2000	2019	1200	1100	1069	1000	0
5	50000	1	2	1	57	-1	0	-1	0	0	0	8617	5670	35835	20940	19146	19131	2000	36681	10000	9000	689	679	0
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
29996	220000	1	3	1	39	0	0	0	0	0	0	188948	192815	208365	88004	31237	15980	8500	20000	5003	3047	5000	1000	0
29997	150000	1	3	2	43	-1	-1	-1	-1	0	0	1683	1828	3502	8979	5190	0	1837	3526	8998	129	0	0	0
29998	30000	1	2	2	37	4	3	2	-1	0	0	3565	3356	2758	20878	20582	19357	0	0	22000	4200	2000	3100	1
29999	80000	1	3	1	41	1	-1	0	0	0	-1	-1645	78379	76304	52774	11855	48944	85900	3409	1178	1926	52964	1804	1
30000	50000	1	2	1	46	0	0	0	0	0	0	47929	48905	49764	36535	32428	15313	2078	1800	1430	1000	1000	1000	1

30000 rows × 24 columns

```
<class 'pandas.DataFrame'>
RangeIndex: 30000 entries, 1 to 30000
Data columns (total 24 columns):
 #   Column                      Non-Null Count  Dtype
---  ------                      --------------  -----
 0   LIMIT_BAL                   30000 non-null  int64
 1   SEX                         30000 non-null  int64
 2   EDUCATION                   30000 non-null  int64
 3   MARRIAGE                    30000 non-null  int64
 4   AGE                         30000 non-null  int64
 5   PAY_0                       30000 non-null  int64
 6   PAY_2                       30000 non-null  int64
 7   PAY_3                       30000 non-null  int64
 8   PAY_4                       30000 non-null  int64
 9   PAY_5                       30000 non-null  int64
 10  PAY_6                       30000 non-null  int64
 11  BILL_AMT1                   30000 non-null  float64
 12  BILL_AMT2                   30000 non-null  float64
 13  BILL_AMT3                   30000 non-null  float64
 14  BILL_AMT4                   30000 non-null  float64
 15  BILL_AMT5                   30000 non-null  float64
 16  BILL_AMT6                   30000 non-null  float64
 17  PAY_AMT1                    30000 non-null  float64
 18  PAY_AMT2                    30000 non-null  float64
 19  PAY_AMT3                    30000 non-null  float64
 20  PAY_AMT4                    30000 non-null  float64
 21  PAY_AMT5                    30000 non-null  float64
 22  PAY_AMT6                    30000 non-null  float64
 23  default payment next month  30000 non-null  int64
dtypes: float64(12), int64(12)
memory usage: 5.5 MB

Index(['LIMIT_BAL', 'SEX', 'EDUCATION', 'MARRIAGE', 'AGE', 'PAY_0',
'PAY_2',
       'PAY_3', 'PAY_4', 'PAY_5', 'PAY_6', 'BILL_AMT1', 'BILL_AMT2',
       'BILL_AMT3', 'BILL_AMT4', 'BILL_AMT5', 'BILL_AMT6', 'PAY_AMT1',
       'PAY_AMT2', 'PAY_AMT3', 'PAY_AMT4', 'PAY_AMT5', 'PAY_AMT6',
       'default payment next month'],
      dtype='object', name=0)
```

	LIMIT_BAL	SEX	EDUCATION	MARRIAGE	AGE	PAY_0	PAY_2	PAY_3	PAY_4	PAY_5	PAY_6	BILL_AMT1	BILL_AMT2	BILL_AMT3	BILL_AMT4	BILL_AMT5	BILL_AMT6	PAY_AMT1	PAY_AMT2	PAY_AMT3	PAY_AMT4	PAY_AMT5	PAY_AMT6	default payment next month	[illegible]
1	20000	2	2	1	24	2	2	-1	-1	-2	-2	3913.0	3102.0	689.0	0.0	0.0	0.0	0.0	689.0	0.0	0.0	0.0	0.0	1	7015.0
2	120000	2	2	2	26	-1	2	0	0	0	2	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	1000.0	[illegible]	[illegible]	[illegible]	2000.0	1	12077
3	90000	2	2	2	34	0	0	0	0	0	0	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	0	93671
4	50000	2	2	1	37	0	0	0	0	0	0	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	0	22694
5	50000	1	2	1	57	-1	0	-1	0	0	0	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	679.0	0	54290
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
29996	220000	1	3	1	39	0	0	0	0	0	0	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	0	69979
29997	150000	1	3	2	43	-1	-1	-1	-1	0	0	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	0.0	[illegible]	[illegible]	[illegible]	[illegible]	0.0	0.0	0	10366
29998	30000	1	2	2	37	4	3	2	-1	0	0	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	0.0	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	1	39196
29999	80000	1	3	1	41	1	-1	0	0	0	-1	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	1	29123
30000	50000	1	2	1	46	0	0	0	0	0	0	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	1	22672

30000 rows × 25 columns

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         480000,  350000,  330000,  110000,  420000,  170000,  370000,
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         490000,  390000,  580000,  600000,  620000,  610000,  700000,
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 'EDUCATION': array([2, 1, 3, 4]),
 'MARRIAGE': array([1, 2, 3]),
 'AGE': array([24, 26, 34, 37, 57, 29, 23, 28, 35, 51, 41, 30, 49, 39, 40,
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36,
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65,
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 'REPAY_STATUS_JUNE': array([-1,  0, -2,  2,  3,  4,  5,  7,  6,  1,  8]),
 'REPAY_STATUS_MAY': array([-2,  0, -1,  2,  3,  5,  4,  7,  8,  6]),
 'REPAY_STATUS_APR': array([-2,  2,  0, -1,  3,  6,  4,  7,  8,  5]),
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       shape=(21010,)),
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```
8049.], shape=(6937,)),
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52964.], shape=(6897,)),
 'PAY_AMT_APR': array([     0.,   2000.,   5000., ...,  70052., 220076.,
16080.],
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 'default payment next month': array([1, 0]),
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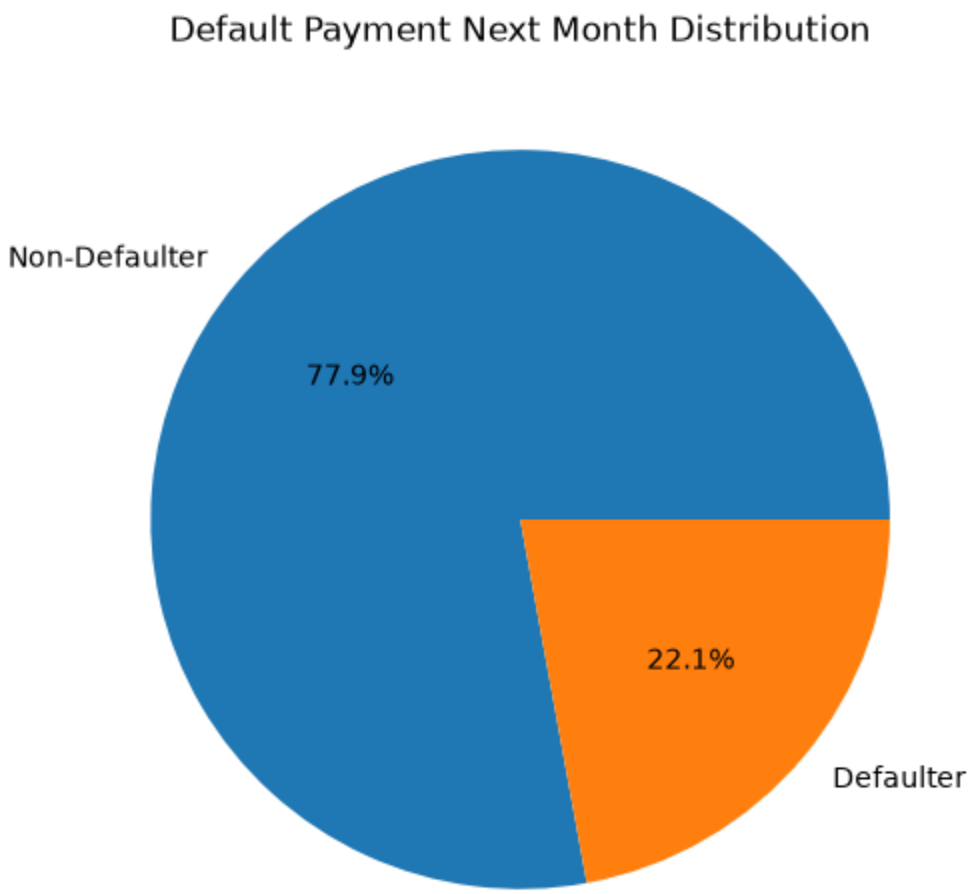
1	20000	2	2	1	24	2	2	-1	-1	-2	-2	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
2	120000	2	2	2	26	-1	2	0	0	0	2	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
3	90000	2	2	2	34	0	0	0	0	0	0	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
4	50000	2	2	1	37	0	0	0	0	0	0	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
5	50000	1	2	1	57	-1	0	-1	0	0	0	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
29996	220000	1	3	1	39	0	0	0	0	0	0	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
29997	150000	1	3	2	43	-1	-1	-1	-1	0	0	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
29998	30000	1	2	2	37	4	3	2	-1	0	0	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
29999	80000	1	3	1	41	1	-1	0	0	0	-1	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
30000	50000	1	2	1	46	0	0	0	0	0	0	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]

30000 rows × 25 columns

	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	default payment next month	[illegible]
count	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
mean	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
std	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
min	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
25%	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
50%	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
75%	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
max	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]

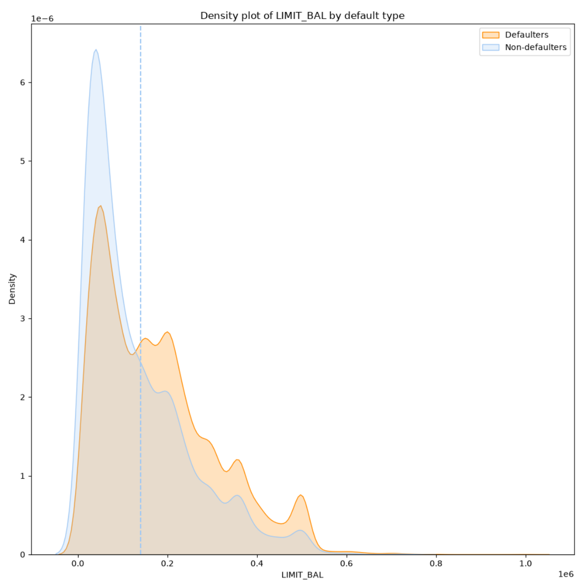
EDA

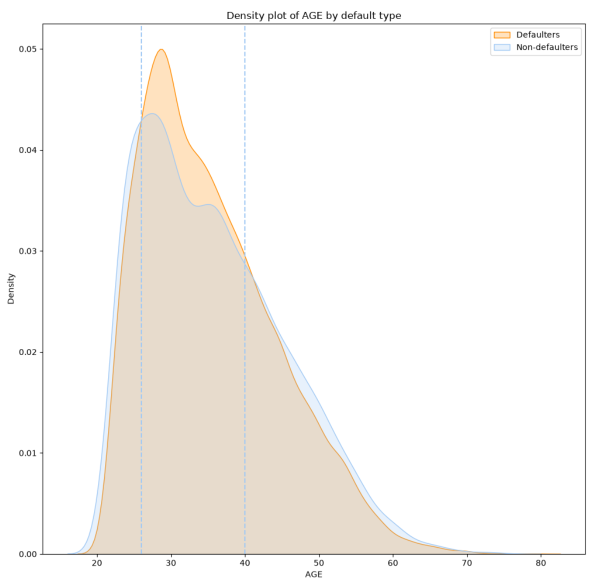


The Pie chart represents the percentage of customer who will be either Non-Defaulter or Defaulter Next Month.

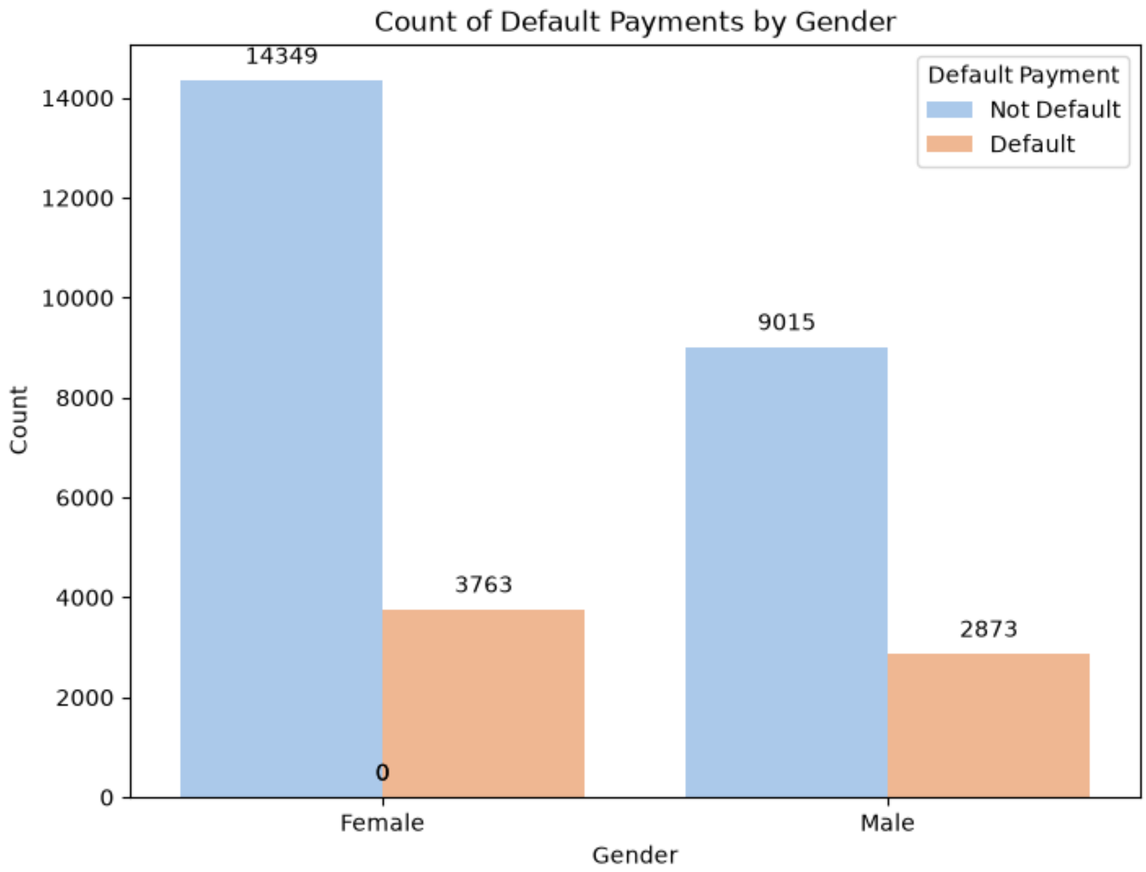
It is visible that majority of the customers will be Non-defaulter the next month with 77.9%.

22.1% of the customers might be defaulter next month

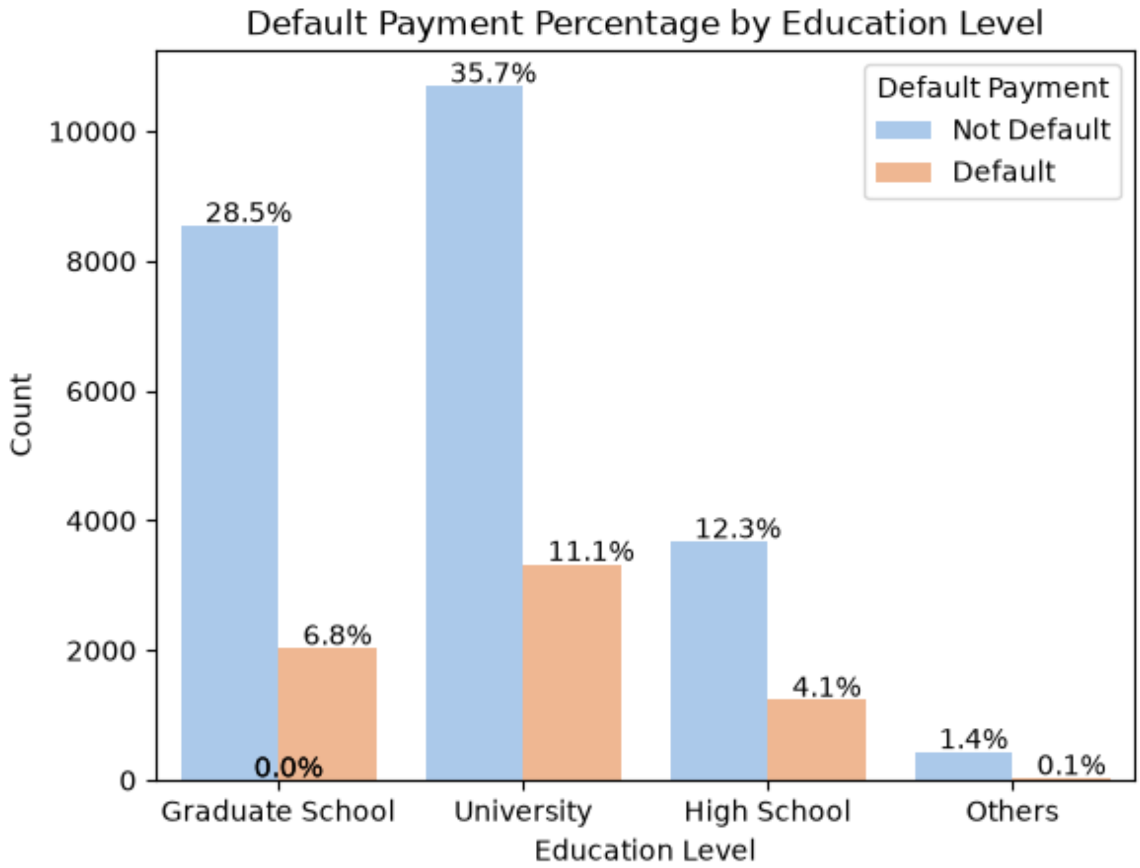




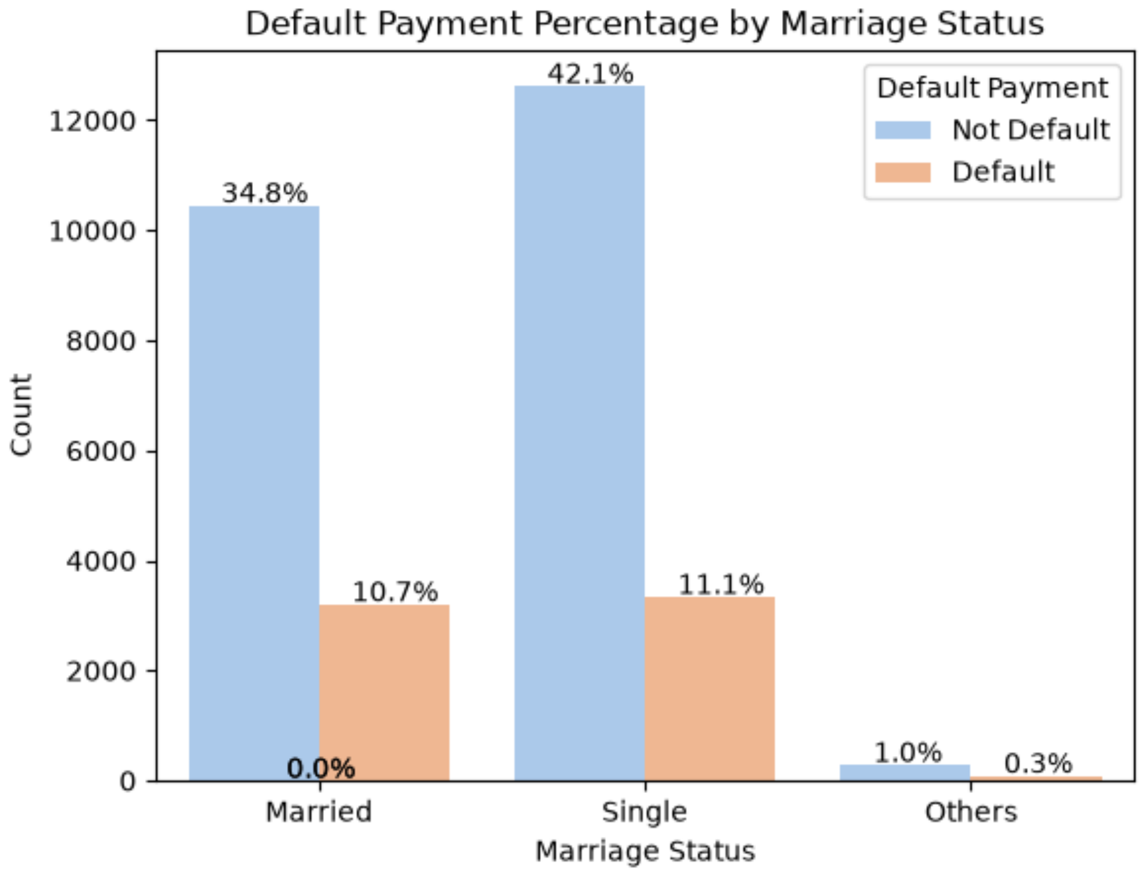
The above graph shows denisty plot of Defaulter Type vs LIMIT_BAL and Age vs Defaulter Type. Users having low credit car limits tends to be non-defaulters. As the limit increases the probability of beiing defaulter increases. Approximately From ages 25 - 40 had highest number of defaulters which started to decrese with increase in age.



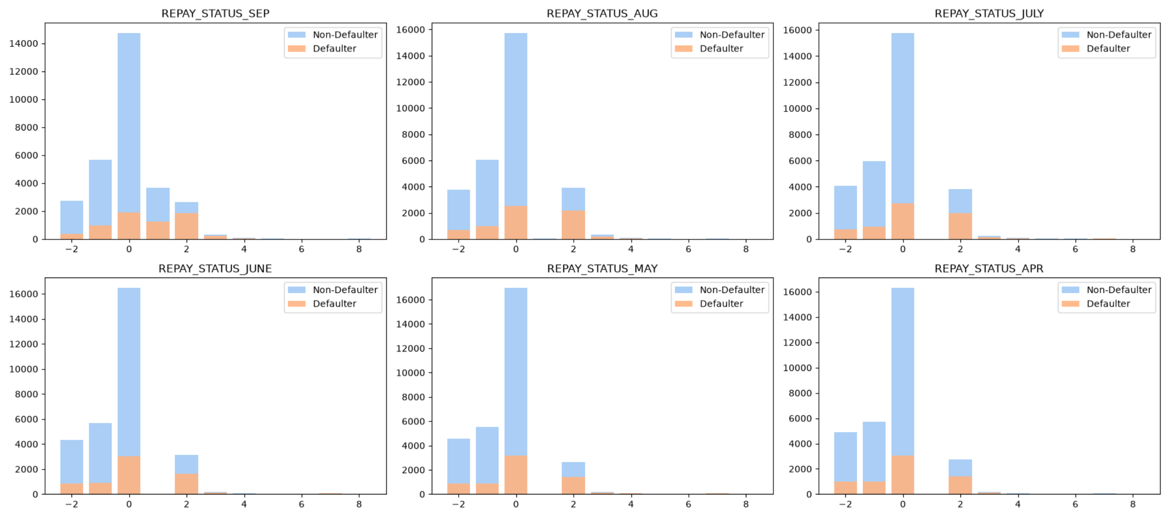
The bar chart represents Default payments by gender. Here it can be seen that the females are majority credit card holders with total of 18112. Males credit card holders are 11888. As majority of credit card users are Females, From the graph it is observed that females are majority defaulters.



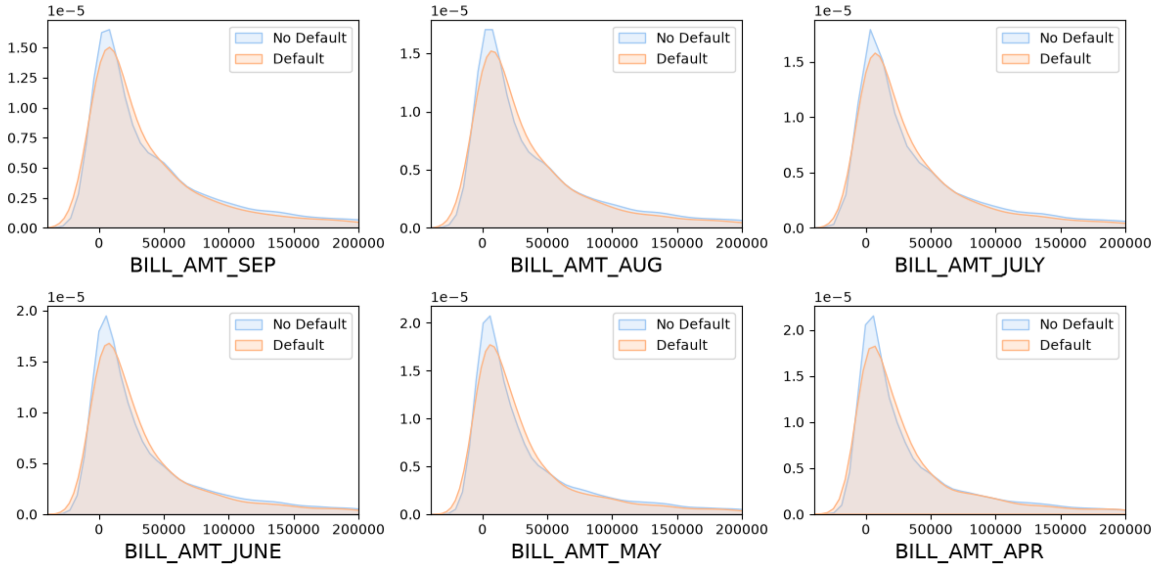
The above graph represents Type of defaulters with respect to their Education level. Here it is observed that Users having University level education are maximum Defaulters while high-school and other category have lowest Defaulters.



The above graph shows Marriage status with respect to Defaulter Type. Here we can see that the single category contributes to majority credit card users as well as majority in Defaulters

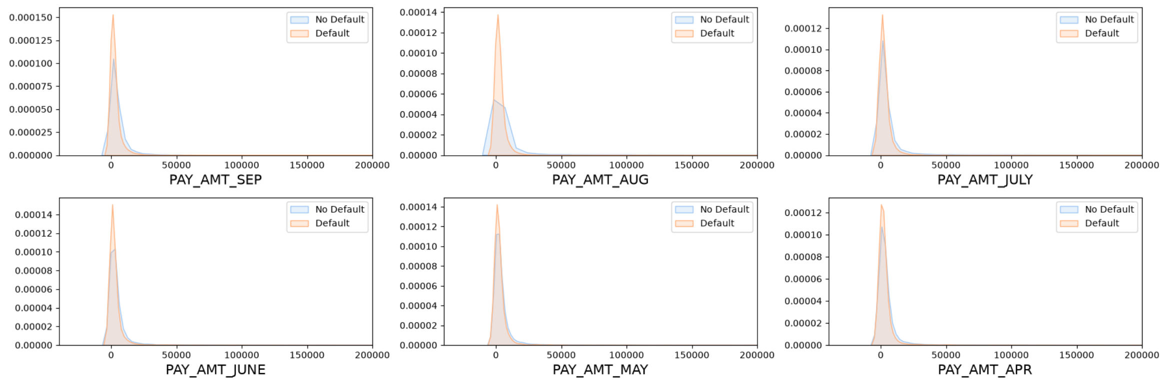


The chart shows us the repay Status for 6 months and their respective counts for Users.



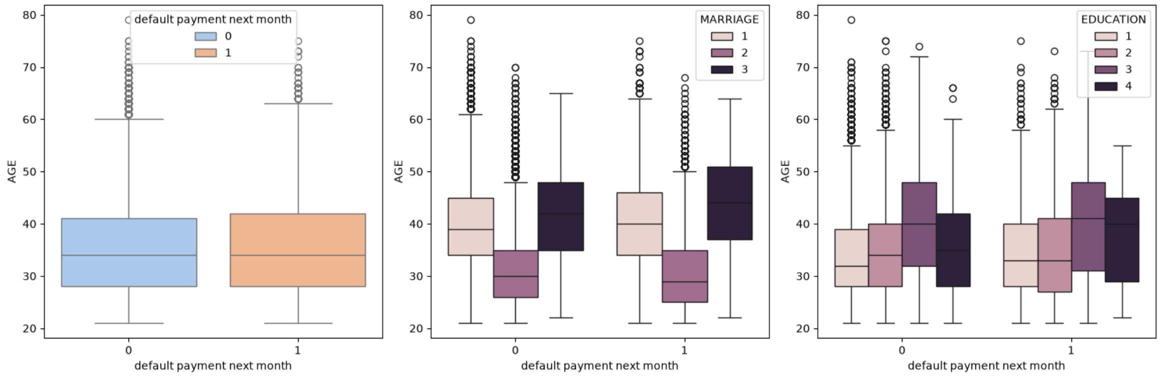
The above graph shows density distributions of Bill amounts for 6 months (April, May, June, July, August, and September) for defaulters and non-defaulters over time.

It is visible that for lower billing amount users tend to be non-defaulters.



The above graph shows density distributions of Pay Amount for 6 months (April, May, June, July, August, September) for defaulters and non-defaulters over the time.

It is visible that for low paying amount users tend to be defaulters.

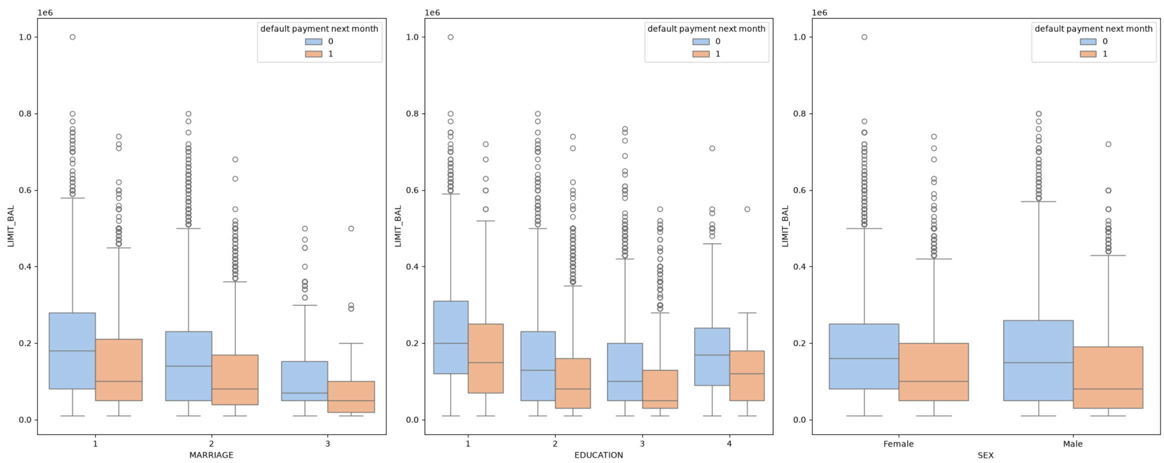


The first graph shows spread of data for default payment next month Vs. Age, here we can see that there are lot of outliers in non-default users.

The next graphs show spread of data for default payment next month Vs. Age with respect to Marriage & Education.

It is visible that Marriage group 2 has the highest number of outliers for both users, while group 3 has no outliers.

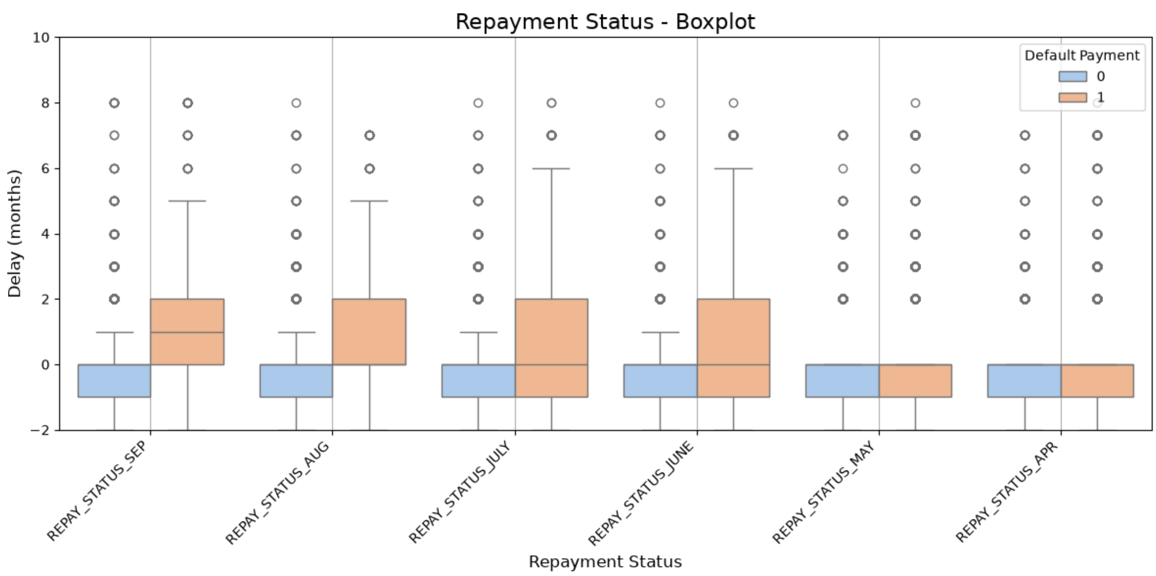
For Education data the median lies between 30 – 40 range with maximum number of outliers in group 1.



These graphs show spread of data for Marriage, Education, Sex Vs. Limit Balance with respect to two user groups in our data.

The position of median for most of the groups suggests that the lower half of the data is more densely packed.

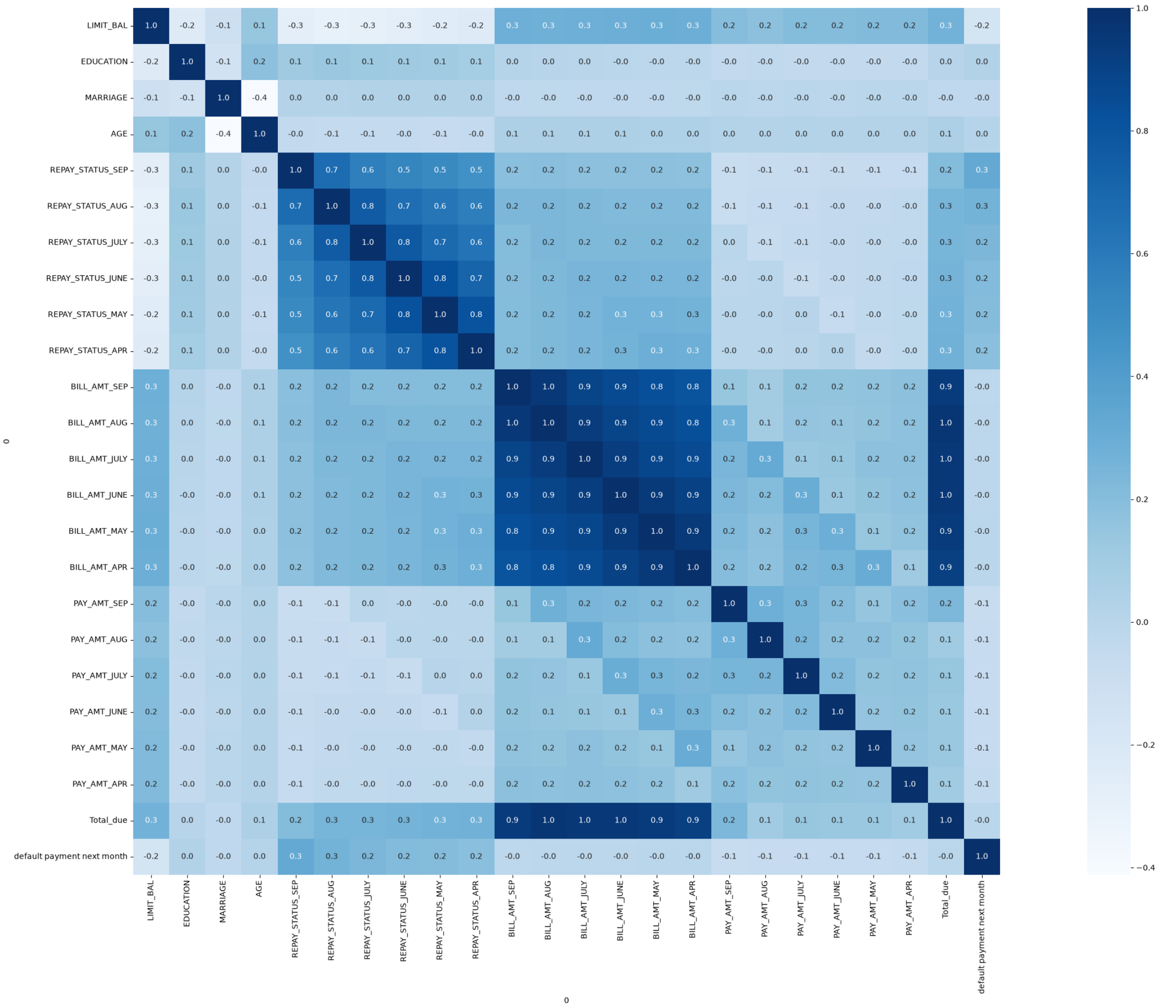
For all three visualizations non-defaulters have the maximum number of outliers.



The above graph shows plots of Repay status for 6 months (April, May, June, July, August, September) with respect to Delay.

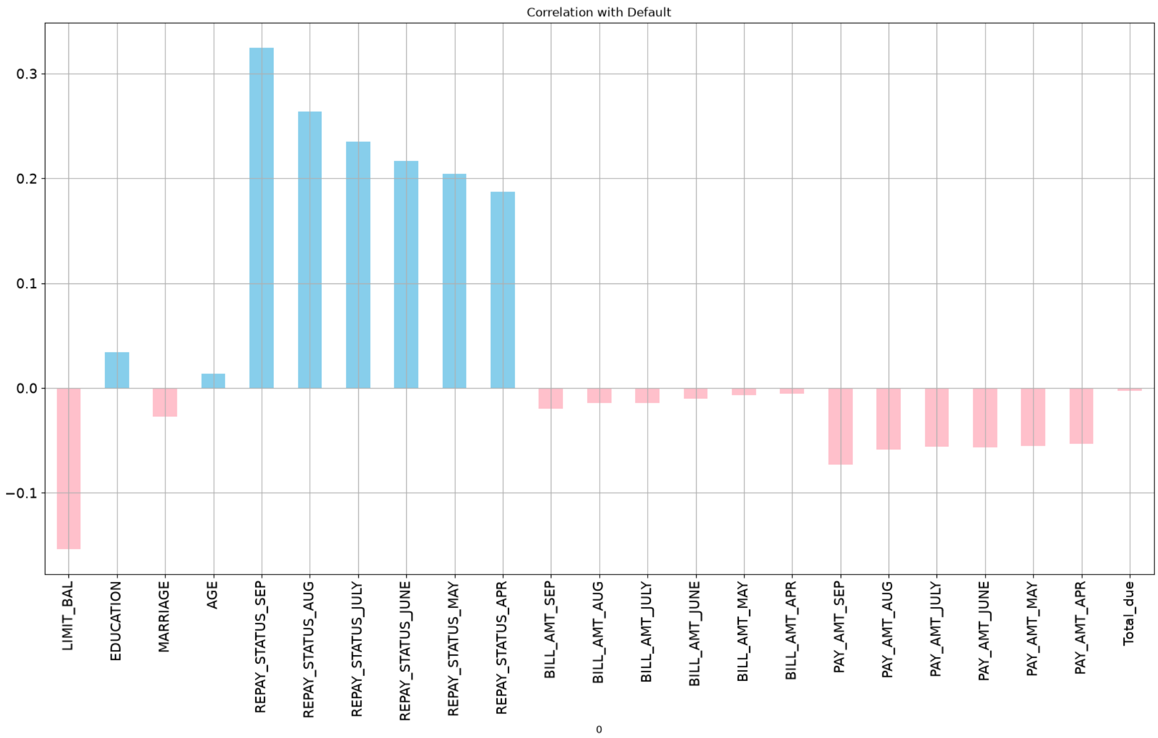
There are more outliers for group 0, with more dense distribution towards lower delay values.

```
<Axes: xlabel='0', ylabel='0'>
```



The heatmap shows visual correlation matrices between all the variables in the data.

Hence, we can easily identify relationships between variables. High or low correlation coefficients are visible by lighter or darker color intensities.



This graph shoes correlation of Default variable with all the variables in data.

Here, we can observe that Repay Status for Sept has the highest positive correlation and Limit_Bal has the highest negative correlation.

Data Transformations

	[illegible]								[illegible] default payment next month
1	[illegible]	2	2	-1	-1	-2	-2	[illegible]	[illegible]
2	[illegible]	-1	2	0	0	0	2	[illegible]	[illegible]
3	[illegible]	0	0	0	0	0	0	[illegible]	[illegible]
4	[illegible]	0	0	0	0	0	0	[illegible]	[illegible]
5	[illegible]	-1	0	-1	0	0	0	[illegible]	[illegible]
...	...	...	...	...	...	...	...	...	...
[illegible]	[illegible]	0	0	0	0	0	0	[illegible]	[illegible]
[illegible]	[illegible]	-1	-1	-1	-1	0	0	[illegible]	[illegible]
[illegible]	[illegible]	4	3	2	-1	0	0	[illegible]	[illegible]
[illegible]	[illegible]	1	-1	0	0	0	-1	[illegible]	[illegible]
[illegible]	[illegible]	0	0	0	0	0	0	[illegible]	[illegible]

30000 rows × 30 columns

1	[illegible]	24	2	2	-1	-1	-2	-2	[illegible]
2	[illegible]	26	-1	2	0	0	0	2	[illegible]
3	[illegible]	34	0	0	0	0	0	0	[illegible]
4	[illegible]	37	0	0	0	0	0	0	[illegible]
5	[illegible]	57	-1	0	-1	0	0	0	[illegible]
6	[illegible]	37	0	0	0	0	0	0	[illegible]
7	[illegible]	29	0	0	0	0	0	0	[illegible]
8	[illegible]	23	0	-1	-1	0	0	-1	[illegible]
9	[illegible]	28	0	0	2	0	0	0	[illegible]
10	[illegible]	35	-2	-2	-2	-2	-1	-1	[illegible]

```
Index(['LIMIT_BAL', 'SEX', 'EDUCATION_4', 'EDUCATION_3', 'EDUCATION_2',
       'EDUCATION_1', 'MARRIAGE_3', 'MARRIAGE_2', 'MARRIAGE_1', 'AGE',
       'REPAY_STATUS_SEP', 'REPAY_STATUS_AUG', 'REPAY_STATUS_JULY',
       'REPAY_STATUS_JUNE', 'REPAY_STATUS_MAY', 'REPAY_STATUS_APR',
       'BILL_AMT_SEP', 'BILL_AMT_AUG', 'BILL_AMT_JULY', 'BILL_AMT_JUNE',
       'BILL_AMT_MAY', 'BILL_AMT_APR', 'PAY_AMT_SEP', 'PAY_AMT_AUG',
       'PAY_AMT_JULY', 'PAY_AMT_JUNE', 'PAY_AMT_MAY', 'PAY_AMT_APR',
       'Total_due', 'default payment next month'],
      dtype='str', name=0)
```

1	[illegible]
2	[illegible]
3	[illegible]
4	[illegible]
5	[illegible]
...	...
29996	[illegible]
29997	[illegible]
29998	[illegible]
29999	[illegible]
30000	[illegible]

30000 rows × 30 columns

```
<class 'pandas.DataFrame'>
RangeIndex: 30000 entries, 1 to 30000
Data columns (total 30 columns):
 #   Column                      Non-Null Count  Dtype
---  ------                      --------------  -----
 0   LIMIT_BAL                   30000 non-null  float64
 1   SEX                         30000 non-null  int64
 2   EDUCATION_4                 30000 non-null  bool
 3   EDUCATION_3                 30000 non-null  bool
 4   EDUCATION_2                 30000 non-null  bool
 5   EDUCATION_1                 30000 non-null  bool
 6   MARRIAGE_3                  30000 non-null  bool
 7   MARRIAGE_2                  30000 non-null  bool
 8   MARRIAGE_1                  30000 non-null  bool
 9   AGE                         30000 non-null  float64
 10  REPAY_STATUS_SEP            30000 non-null  float64
 11  REPAY_STATUS_AUG            30000 non-null  float64
 12  REPAY_STATUS_JULY           30000 non-null  float64
 13  REPAY_STATUS_JUNE           30000 non-null  float64
 14  REPAY_STATUS_MAY            30000 non-null  float64
 15  REPAY_STATUS_APR            30000 non-null  float64
 16  BILL_AMT_SEP                30000 non-null  float64
 17  BILL_AMT_AUG                30000 non-null  float64
 18  BILL_AMT_JULY               30000 non-null  float64
 19  BILL_AMT_JUNE               30000 non-null  float64
 20  BILL_AMT_MAY                30000 non-null  float64
 21  BILL_AMT_APR                30000 non-null  float64
 22  PAY_AMT_SEP                 30000 non-null  float64
 23  PAY_AMT_AUG                 30000 non-null  float64
 24  PAY_AMT_JULY                30000 non-null  float64
 25  PAY_AMT_JUNE                30000 non-null  float64
 26  PAY_AMT_MAY                 30000 non-null  float64
 27  PAY_AMT_APR                 30000 non-null  float64
 28  Total_due                   30000 non-null  float64
 29  default payment next month  30000 non-null  int64
dtypes: bool(7), float64(21), int64(2)
memory usage: 5.5 MB

default payment next month
0    23364
1     6636
Name: count, dtype: int64

(30000, 30)
```

	[illegible]
0	[illegible]
1	[illegible]
2	[illegible]
3	[illegible]
4	[illegible]
...	...
46723	[illegible]
46724	[illegible]
46725	[illegible]
46726	[illegible]
46727	[illegible]

46728 rows × 30 columns

```
default payment next month
1    23364
0    23364
Name: count, dtype: int64

(46728, 30)
```

```
<class 'pandas.DataFrame'>
RangeIndex: 46728 entries, 0 to 46727
Data columns (total 30 columns):
 #   Column                      Non-Null Count  Dtype  
---  ------                      --------------  -----  
 0   LIMIT_BAL                   46728 non-null  float64
 1   SEX                         46728 non-null  int64  
 2   EDUCATION_4                 46728 non-null  bool   
 3   EDUCATION_3                 46728 non-null  bool   
 4   EDUCATION_2                 46728 non-null  bool   
 5   EDUCATION_1                 46728 non-null  bool   
 6   MARRIAGE_3                  46728 non-null  bool   
 7   MARRIAGE_2                  46728 non-null  bool   
 8   MARRIAGE_1                  46728 non-null  bool   
 9   AGE                         46728 non-null  float64
 10  REPAY_STATUS_SEP            46728 non-null  float64
 11  REPAY_STATUS_AUG            46728 non-null  float64
 12  REPAY_STATUS_JULY           46728 non-null  float64
 13  REPAY_STATUS_JUNE           46728 non-null  float64
 14  REPAY_STATUS_MAY            46728 non-null  float64
 15  REPAY_STATUS_APR            46728 non-null  float64
 16  BILL_AMT_SEP                46728 non-null  float64
 17  BILL_AMT_AUG                46728 non-null  float64
 18  BILL_AMT_JULY               46728 non-null  float64
 19  BILL_AMT_JUNE               46728 non-null  float64
 20  BILL_AMT_MAY                46728 non-null  float64
 21  BILL_AMT_APR                46728 non-null  float64
 22  PAY_AMT_SEP                 46728 non-null  float64
 23  PAY_AMT_AUG                 46728 non-null  float64
 24  PAY_AMT_JULY                46728 non-null  float64
 25  PAY_AMT_JUNE                46728 non-null  float64
 26  PAY_AMT_MAY                 46728 non-null  float64
 27  PAY_AMT_APR                 46728 non-null  float64
 28  Total_due                   46728 non-null  float64
 29  default payment next month  46728 non-null  int64  
dtypes: bool(7), float64(21), int64(2)
memory usage: 8.5 MB
```

0	[illegible]
LIMI[illegible]	[illegible]
SEX	[illegible]
EDU[illegible]	[illegible]
EDU[illegible]	[illegible]
EDU[illegible]	[illegible]
EDU[illegible]	[illegible]
MA[illegible]	[illegible]
MA[illegible]	[illegible]
MA[illegible]	[illegible]
AGE	[illegible]
REP[illegible]	[illegible]
REP[illegible]	[illegible]
REP[illegible]	[illegible]
REP[illegible]	[illegible]
REP[illegible]	[illegible]
REP[illegible]	[illegible]
BIL[illegible]	[illegible]
BIL[illegible]	[illegible]
BIL[illegible]	[illegible]
BIL[illegible]	[illegible]
BIL[illegible]	[illegible]
BIL[illegible]	[illegible]
PAY[illegible]	[illegible]
PAY[illegible]	[illegible]
PAY[illegible]	[illegible]
PAY[illegible]	[illegible]
PAY[illegible]	[illegible]
PAY[illegible]	[illegible]
Total[illegible]	[illegible]
default payment next month	-0.2[illegible]

Dataset Splitting

Models Development

```
[notice] A new release of pip is available: 25.3 -> 26.1.2
[notice] To update, run: pip install --upgrade pip
```

Logistic Regression (Base Model)

```
Solving using gradient descent
```

```
  0%|                                                            | 0/5000
[00:00<?, ?it/s]/var/folders/hg/9_9zzq252fsdzz4by4y0lvrw0000gn/T/
ipykernel_4793/2760970217.py:30: RuntimeWarning: overflow encountered in
exp
  return 1 / (1 + np.exp(-z))
100%|████████████████████████████████████████| 5000/5000
[00:02<00:00, 1762.99it/s]
```

```
Precision: 62.157
Recall: 60.893
Accuracy: 62.164
Specificity: 63.418
F1-Score: 61.518
```

```
Bias: 0.379
Variance: 0.250
Noise: 0.378
Expected Loss: 1.007
```

```
peak memory: 537.73 MiB, increment: 3.69 MiB
```

```
Logistic Regression - Elapsed time: 3.208 seconds
```

Hard-Margin SVM

```
Hard-Margin SVM Progress: 100%|█████████████████████| 700/700
[00:34<00:00, 20.35it/s]
```

```
peak memory: 544.94 MiB, increment: 8.08 MiB
peak memory: 524.72 MiB, increment: 0.31 MiB
```

```
Precision: 77.814
Recall: 61.358
Accuracy: 72.120
Specificity: 82.738
F1-Score: 68.613
```

```
Bias: 1.305
Variance: 0.945
Noise: 1.271
Expected Loss: 3.522
```

```
peak memory: 540.25 MiB, increment: 15.84 MiB
```

```
Hard-Margin SVM - Elapsed time: 35.503 seconds
```

Gaussian Naive Bayes

```
Gaussian Naive Bayes Progress: 100%|██████████| 11682/11682
[00:12<00:00, 929.94it/s]
Gaussian Naive Bayes Progress: 100%|██████████| 35046/35046
[00:38<00:00, 919.65it/s]
Gaussian Naive Bayes Progress: 100%|██████████| 35046/35046
[00:38<00:00, 910.82it/s]
Gaussian Naive Bayes Progress: 100%|██████████| 35046/35046
[00:38<00:00, 899.74it/s]
```

```
Precision: 60.113
Recall: 73.095
Accuracy: 62.549
Specificity: 52.143
F1-Score: 65.972
```

```
Bias: 0.376
Variance: 0.239
Noise: 0.375
Expected Loss: 0.990
```

```
peak memory: 548.78 MiB, increment: 0.03 MiB
```

```
Gaussian Naive Bayes - Elapsed time: 128.446 seconds
```

Neural Networks (Multi-Layer Perceptron Classifier using TensorFlow, Keras)

```
Requirement already satisfied: keras in ./.venv/lib/python3.11/site-
packages (3.15.0)
Requirement already satisfied: absl-py in ./.venv/lib/python3.11/site-
packages (from keras) (2.4.0)
Requirement already satisfied: numpy in ./.venv/lib/python3.11/site-
packages (from keras) (2.4.6)
Requirement already satisfied: rich in ./.venv/lib/python3.11/site-
packages (from keras) (15.0.0)
Requirement already satisfied: namex in ./.venv/lib/python3.11/site-
packages (from keras) (0.1.0)
Requirement already satisfied: h5py in ./.venv/lib/python3.11/site-
packages (from keras) (3.14.0)
Requirement already satisfied: optree in ./.venv/lib/python3.11/site-
packages (from keras) (0.19.1)
Requirement already satisfied: ml-dtypes in ./.venv/lib/python3.11/site-
packages (from keras) (0.5.4)
Requirement already satisfied: packaging in ./.venv/lib/python3.11/site-
packages (from keras) (26.2)
Requirement already satisfied: typing-extensions>=4.6.0 in ./.venv/lib/
python3.11/site-packages (from optree->keras) (4.15.0)
Requirement already satisfied: markdown-it-py>=2.2.0 in ./.venv/lib/
python3.11/site-packages (from rich->keras) (4.2.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in ./.venv/lib/
python3.11/site-packages (from rich->keras) (2.20.0)
Requirement already satisfied: mdurl~=0.1 in ./.venv/lib/python3.11/site-
packages (from markdown-it-py>=2.2.0->rich->keras) (0.1.2)

[notice] A new release of pip is available: 25.3 -> 26.1.2
[notice] To update, run: pip install --upgrade pip
Requirement already satisfied: tensorflow in ./.venv/lib/python3.11/site-
packages (2.21.0)
Requirement already satisfied: absl-py>=1.0.0 in ./.venv/lib/python3.11/
site-packages (from tensorflow) (2.4.0)
Requirement already satisfied: astunparse>=1.6.0 in ./.venv/lib/
python3.11/site-packages (from tensorflow) (1.6.3)
Requirement already satisfied: flatbuffers>=25.9.23 in ./.venv/lib/
python3.11/site-packages (from tensorflow) (25.12.19)
Requirement already satisfied: gast!=0.5.0,!=0.5.1,!=0.5.2,>=0.2.1
in ./.venv/lib/python3.11/site-packages (from tensorflow) (0.7.0)
Requirement already satisfied: google_pasta>=0.1.1 in ./.venv/lib/
python3.11/site-packages (from tensorflow) (0.2.0)
Requirement already satisfied: libclang>=13.0.0 in ./.venv/lib/python3.11/
site-packages (from tensorflow) (18.1.1)
Requirement already satisfied: opt_einsum>=2.3.2 in ./.venv/lib/
python3.11/site-packages (from tensorflow) (3.4.0)
Requirement already satisfied: packaging in ./.venv/lib/python3.11/site-
packages (from tensorflow) (26.2)
Requirement already satisfied: protobuf<8.0.0,>=6.31.1 in ./.venv/lib/
python3.11/site-packages (from tensorflow) (7.35.1)
Requirement already satisfied: requests<3,>=2.21.0 in ./.venv/lib/
python3.11/site-packages (from tensorflow) (2.34.2)
Requirement already satisfied: setuptools in ./.venv/lib/python3.11/site-
packages (from tensorflow) (80.9.0)
Requirement already satisfied: six>=1.12.0 in ./.venv/lib/python3.11/site-
packages (from tensorflow) (1.17.0)
Requirement already satisfied: termcolor>=1.1.0 in ./.venv/lib/python3.11/
```

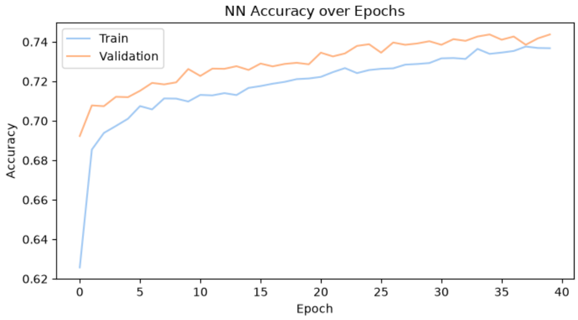
```
site-packages (from tensorflow) (3.3.0)
Requirement already satisfied: typing_extensions>=3.6.6 in ./.venv/lib/
python3.11/site-packages (from tensorflow) (4.15.0)
Requirement already satisfied: wrapt>=1.11.0 in ./.venv/lib/python3.11/
site-packages (from tensorflow) (2.2.2)
Requirement already satisfied: grpcio<2.0,>=1.24.3 in ./.venv/lib/
python3.11/site-packages (from tensorflow) (1.81.1)
Requirement already satisfied: keras>=3.12.0 in ./.venv/lib/python3.11/
site-packages (from tensorflow) (3.15.0)
Requirement already satisfied: numpy>=1.26.0 in ./.venv/lib/python3.11/
site-packages (from tensorflow) (2.4.6)
Requirement already satisfied: h5py<3.15.0,>=3.11.0 in ./.venv/lib/
python3.11/site-packages (from tensorflow) (3.14.0)
Requirement already satisfied: ml_dtypes<1.0.0,>=0.5.1 in ./.venv/lib/
python3.11/site-packages (from tensorflow) (0.5.4)
Requirement already satisfied: charset_normalizer<4,>=2 in ./.venv/lib/
python3.11/site-packages (from requests<3,>=2.21.0->tensorflow) (3.4.7)
Requirement already satisfied: idna<4,>=2.5 in ./.venv/lib/python3.11/
site-packages (from requests<3,>=2.21.0->tensorflow) (3.18)
Requirement already satisfied: urllib3<3,>=1.26 in ./.venv/lib/python3.11/
site-packages (from requests<3,>=2.21.0->tensorflow) (2.7.0)
Requirement already satisfied: certifi>=2023.5.7 in ./.venv/lib/
python3.11/site-packages (from requests<3,>=2.21.0->tensorflow)
(2026.6.17)
Requirement already satisfied: wheel<1.0,>=0.23.0 in ./.venv/lib/
python3.11/site-packages (from astunparse>=1.6.0->tensorflow) (0.47.0)
Requirement already satisfied: rich in ./.venv/lib/python3.11/site-
packages (from keras>=3.12.0->tensorflow) (15.0.0)
Requirement already satisfied: namex in ./.venv/lib/python3.11/site-
packages (from keras>=3.12.0->tensorflow) (0.1.0)
Requirement already satisfied: optree in ./.venv/lib/python3.11/site-
packages (from keras>=3.12.0->tensorflow) (0.19.1)
Requirement already satisfied: markdown-it-py>=2.2.0 in ./.venv/lib/
python3.11/site-packages (from rich->keras>=3.12.0->tensorflow) (4.2.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in ./.venv/lib/
python3.11/site-packages (from rich->keras>=3.12.0->tensorflow) (2.20.0)
Requirement already satisfied: mdurl~=0.1 in ./.venv/lib/python3.11/site-
packages (from markdown-it-py>=2.2.0->rich->keras>=3.12.0->tensorflow)
(0.1.2)

[notice] A new release of pip is available: 25.3 -> 26.1.2
[notice] To update, run: pip install --upgrade pip
```

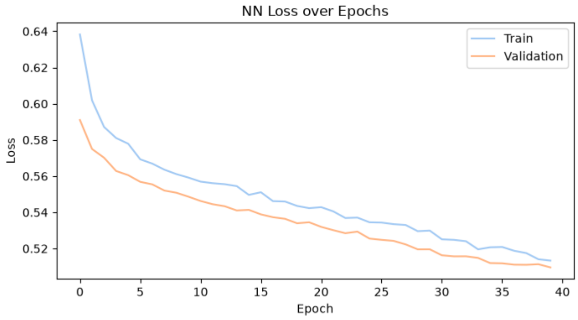
```
/Users/amitojsinghkohli/Documents/Northeastern University/
Semester-3/7300_Machine Learning/Project/Final/.venv/lib/python3.11/site-
packages/keras/src/layers/core/dense.py:107: UserWarning: Do not pass an
`input_shape`/`input_dim` argument to a layer. When using Sequential
models, prefer using an `Input(shape)` object as the first layer in the
model instead.
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
```

```
Epoch 1/40
123/123 ━━━━━━━━━━━━━━━━━━━━
1s 1ms/step - accuracy: 0.6258 - loss: 0.6383 - val_accuracy: 0.6923 -
val_loss: 0.5910
Epoch 2/40
123/123 ━━━━━━━━━━━━━━━━━━━━ 0s 590us/step - accuracy: 0.6854 - loss:
0.6019 - val_accuracy: 0.7078 - val_loss: 0.5750
Epoch 3/40
123/123 ━━━━━━━━━━━━━━━━━━━━ 0s 567us/step - accuracy: 0.6939 - loss:
0.5872 - val_accuracy: 0.7074 - val_loss: 0.5702
Epoch 4/40
123/123 ━━━━━━━━━━━━━━━━━━━━ 0s 566us/step - accuracy: 0.6974 - loss:
0.5811 - val_accuracy: 0.7122 - val_loss: 0.5629
Epoch 5/40
123/123 ━━━━━━━━━━━━━━━━━━━━ 0s 573us/step - accuracy: 0.7011 - loss:
0.5779 - val_accuracy: 0.7120 - val_loss: 0.5605
Epoch 6/40
123/123 ━━━━━━━━━━━━━━━━━━━━ 0s 581us/step - accuracy: 0.7075 - loss:
0.5693 - val_accuracy: 0.7152 - val_loss: 0.5569
Epoch 7/40
123/123 ━━━━━━━━━━━━━━━━━━━━ 0s 619us/step - accuracy: 0.7058 - loss:
0.5669 - val_accuracy: 0.7192 - val_loss: 0.5554
Epoch 8/40
123/123 ━━━━━━━━━━━━━━━━━━━━ 0s 612us/step - accuracy: 0.7114 - loss:
0.5636 - val_accuracy: 0.7185 - val_loss: 0.5520
Epoch 9/40
123/123 ━━━━━━━━━━━━━━━━━━━━ 0s 581us/step - accuracy: 0.7113 - loss:
0.5611 - val_accuracy: 0.7195 - val_loss: 0.5508
Epoch 10/40
123/123 ━━━━━━━━━━━━━━━━━━━━ 0s 564us/step - accuracy: 0.7098 - loss:
0.5592 - val_accuracy: 0.7263 - val_loss: 0.5486
Epoch 11/40
123/123 ━━━━━━━━━━━━━━━━━━━━ 0s 566us/step - accuracy: 0.7132 - loss:
0.5570 - val_accuracy: 0.7228 - val_loss: 0.5463
Epoch 12/40
123/123 ━━━━━━━━━━━━━━━━━━━━ 0s 644us/step - accuracy: 0.7129 - loss:
0.5561 - val_accuracy: 0.7265 - val_loss: 0.5445
Epoch 13/40
123/123 ━━━━━━━━━━━━━━━━━━━━ 0s 580us/step - accuracy: 0.7140 - loss:
0.5556 - val_accuracy: 0.7264 - val_loss: 0.5434
Epoch 14/40
123/123 ━━━━━━━━━━━━━━━━━━━━ 0s 576us/step - accuracy: 0.7131 - loss:
0.5545 - val_accuracy: 0.7277 - val_loss: 0.5410
Epoch 15/40
123/123 ━━━━━━━━━━━━━━━━━━━━ 0s 569us/step - accuracy: 0.7167 - loss:
0.5497 - val_accuracy: 0.7258 - val_loss: 0.5414
Epoch 16/40
123/123 ━━━━━━━━━━━━━━━━━━━━ 0s 566us/step - accuracy: 0.7176 - loss:
0.5511 - val_accuracy: 0.7290 - val_loss: 0.5389
Epoch 17/40
123/123 ━━━━━━━━━━━━━━━━━━━━ 0s 568us/step - accuracy: 0.7188 - loss:
0.5462 - val_accuracy: 0.7276 - val_loss: 0.5373
Epoch 18/40
123/123 ━━━━━━━━━━━━━━━━━━━━ 0s 595us/step - accuracy: 0.7198 - loss:
0.5460 - val_accuracy: 0.7288 - val_loss: 0.5364
Epoch 19/40
123/123 ━━━━━━━━━━━━━━━━━━━━ 0s 591us/step - accuracy: 0.7211 - loss:
```

```
0.5436 - val_accuracy: 0.7294 - val_loss: 0.5340
Epoch 20/40
123/123 ━━━━━━━━━━━━━━━━━━━━ 0s 589us/step - accuracy: 0.7215 - loss: 
0.5423 - val_accuracy: 0.7286 - val_loss: 0.5345
Epoch 21/40
123/123 ━━━━━━━━━━━━━━━━━━━━ 0s 581us/step - accuracy: 0.7223 - loss: 
0.5429 - val_accuracy: 0.7345 - val_loss: 0.5320
Epoch 22/40
123/123 ━━━━━━━━━━━━━━━━━━━━ 0s 591us/step - accuracy: 0.7247 - loss: 
0.5405 - val_accuracy: 0.7326 - val_loss: 0.5302
Epoch 23/40
123/123 ━━━━━━━━━━━━━━━━━━━━ 0s 600us/step - accuracy: 0.7267 - loss: 
0.5369 - val_accuracy: 0.7342 - val_loss: 0.5285
Epoch 24/40
123/123 ━━━━━━━━━━━━━━━━━━━━ 0s 580us/step - accuracy: 0.7242 - loss: 
0.5371 - val_accuracy: 0.7380 - val_loss: 0.5294
Epoch 25/40
123/123 ━━━━━━━━━━━━━━━━━━━━ 0s 581us/step - accuracy: 0.7257 - loss: 
0.5345 - val_accuracy: 0.7388 - val_loss: 0.5255
Epoch 26/40
123/123 ━━━━━━━━━━━━━━━━━━━━ 0s 589us/step - accuracy: 0.7264 - loss: 
0.5344 - val_accuracy: 0.7345 - val_loss: 0.5248
Epoch 27/40
123/123 ━━━━━━━━━━━━━━━━━━━━ 0s 575us/step - accuracy: 0.7266 - loss: 
0.5335 - val_accuracy: 0.7397 - val_loss: 0.5242
Epoch 28/40
123/123 ━━━━━━━━━━━━━━━━━━━━ 0s 589us/step - accuracy: 0.7285 - loss: 
0.5330 - val_accuracy: 0.7385 - val_loss: 0.5223
Epoch 29/40
123/123 ━━━━━━━━━━━━━━━━━━━━ 0s 604us/step - accuracy: 0.7288 - loss: 
0.5296 - val_accuracy: 0.7392 - val_loss: 0.5196
Epoch 30/40
123/123 ━━━━━━━━━━━━━━━━━━━━ 0s 596us/step - accuracy: 0.7293 - loss: 
0.5299 - val_accuracy: 0.7403 - val_loss: 0.5196
Epoch 31/40
123/123 ━━━━━━━━━━━━━━━━━━━━ 0s 591us/step - accuracy: 0.7317 - loss: 
0.5251 - val_accuracy: 0.7385 - val_loss: 0.5163
Epoch 32/40
123/123 ━━━━━━━━━━━━━━━━━━━━ 0s 651us/step - accuracy: 0.7318 - loss: 
0.5248 - val_accuracy: 0.7414 - val_loss: 0.5157
Epoch 33/40
123/123 ━━━━━━━━━━━━━━━━━━━━ 0s 585us/step - accuracy: 0.7314 - loss: 
0.5240 - val_accuracy: 0.7405 - val_loss: 0.5157
Epoch 34/40
123/123 ━━━━━━━━━━━━━━━━━━━━ 0s 582us/step - accuracy: 0.7365 - loss: 
0.5196 - val_accuracy: 0.7427 - val_loss: 0.5148
Epoch 35/40
123/123 ━━━━━━━━━━━━━━━━━━━━ 0s 576us/step - accuracy: 0.7339 - loss: 
0.5207 - val_accuracy: 0.7438 - val_loss: 0.5120
Epoch 36/40
123/123 ━━━━━━━━━━━━━━━━━━━━ 0s 572us/step - accuracy: 0.7346 - loss: 
0.5209 - val_accuracy: 0.7411 - val_loss: 0.5118
Epoch 37/40
123/123 ━━━━━━━━━━━━━━━━━━━━ 0s 582us/step - accuracy: 0.7354 - loss: 
0.5188 - val_accuracy: 0.7427 - val_loss: 0.5111
Epoch 38/40
123/123 ━━━━━━━━━━━━━━━━━━━━ 0s 589us/step - accuracy: 0.7376 - loss: 
```

```
0.5175 - val_accuracy: 0.7385 - val_loss: 0.5110
Epoch 39/40
123/123 ━━━━━━━━━━━━━━━━━━━━ 0s 600us/step - accuracy: 0.7369 - loss:
0.5141 - val_accuracy: 0.7418 - val_loss: 0.5114
Epoch 40/40
123/123 ━━━━━━━━━━━━━━━━━━━━ 0s 592us/step - accuracy: 0.7368 - loss:
0.5133 - val_accuracy: 0.7438 - val_loss: 0.5096
peak memory: 830.38 MiB, increment: 31.09 MiB
```






```
peak memory: 830.81 MiB, increment: 0.00 MiB
366/366 ━━━━━━━━━━━━━━━━━━━━ 0s 197us/step
Accuracy:    75.073
Precision:   77.367
Recall:      70.407
Specificity: 79.677
F1 Score:    73.723
```

```
1096/1096 ━━━━━━━━━━━━━━━━━━━━ 0s 161us/step
1096/1096 ━━━━━━━━━━━━━━━━━━━━ 0s 159us/step
366/366 ━━━━━━━━━━━━━━━━━━━━ 0s 166us/step

Bias: 0.167
Variance: 0.070
Noise: 0.167
Expected Loss: 0.405
```

```
Neural Networks - MLP Classifier - Elapsed time: 5.409 seconds
```

Performance Evaluation and Model Selection

Metrics Comparison:

```
Requirement already satisfied: tabulate in ./.venv/lib/python3.11/site-
packages (0.10.0)

[notice] A new release of pip is available: 25.3 -> 26.1.2
[notice] To update, run: pip install --upgrade pip
Note: you may need to restart the kernel to use updated packages.
```

Model	Precision	Recall	Accuracy	Specificity	F1-Score
Logistic Regression (Base Model)	62.157	60.893	62.164	63.418	61.518
Hard-Margin SVM	77.814	61.358	72.12	82.738	68.613
Gaussian Naive Bayes	60.113	73.095	62.549	52.143	65.972
Neural Networks - MLP Classifier	77.367	70.407	75.073	79.677	73.723

Bias Variance Trade-Off

Model	Bias	Variance	Noise	Expected Loss
Logistic Regression (Base Model)	0.379	0.25	0.378	1.007
Hard-Margin SVM	1.305	0.945	1.271	3.522
Gaussian Naive Bayes	0.376	0.239	0.375	0.99
Neural Networks - MLP Classifier	0.167	0.07	0.167	0.405

Time Complexity

Model	Time (seconds)
Logistic Regression (Base Model)	3.20845
Hard-Margin SVM	35.5033
Gaussian Naive Bayes	128.446
Neural Networks - MLP Classifier	5.40943

Space Complexity

Model	Space (MiB's)
Logistic Regression (Base Model)	371.27
Hard-Margin SVM	371.35
Gaussian Naive Bayes	371.46
Neural Networks - MLP Classifier	807.5

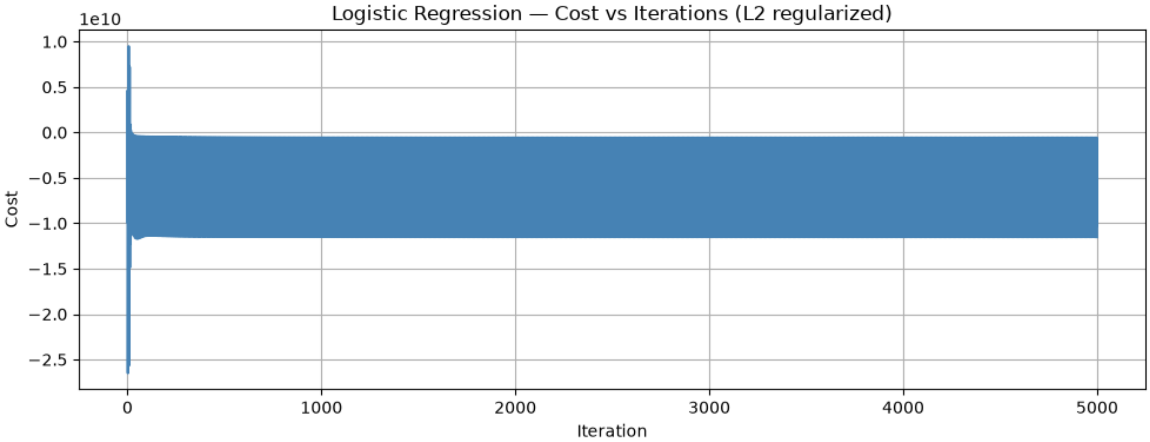
Best Model:

Neural Networks - MultiLayer Perceptron Classifier has been chosen as the best model because it has:

- High precision and specificity values
- Lowest bias, variance and noise
- Execution time is very less
- Good accuracy and recall values

Improvements

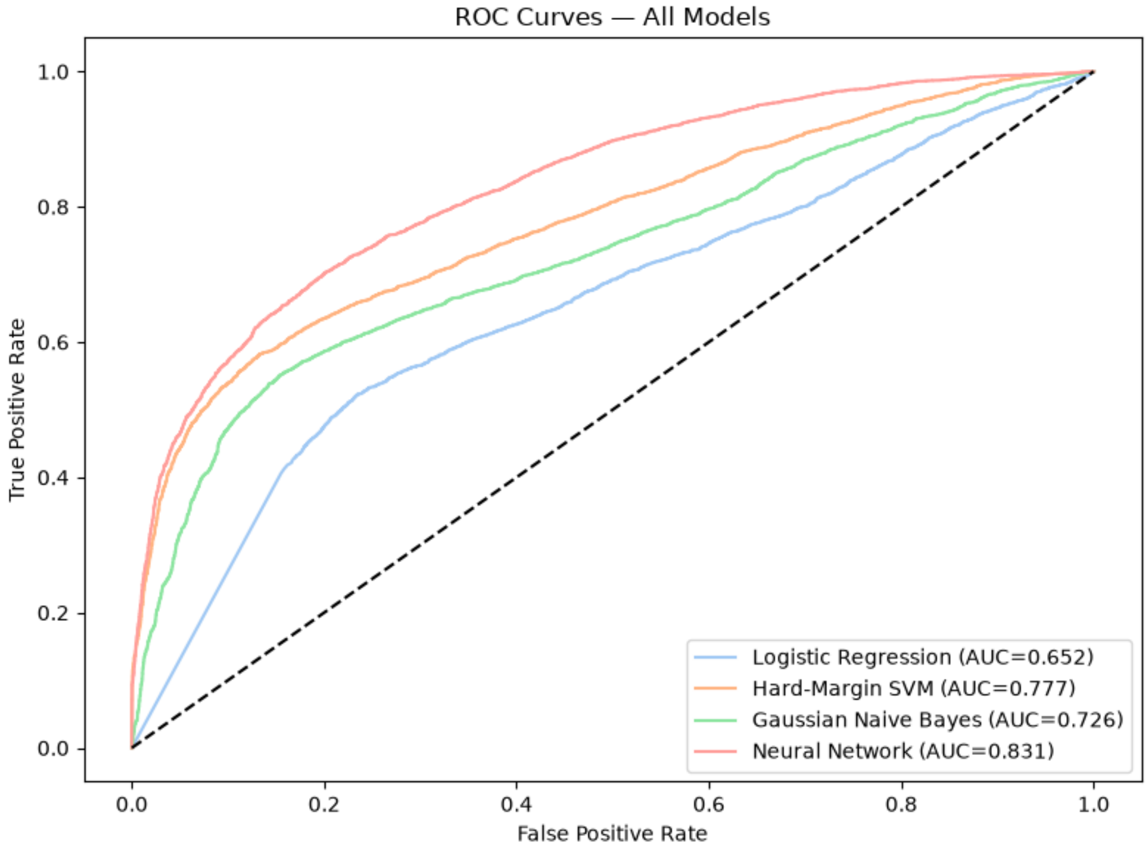
1. Logistic Regression — Cost Curve



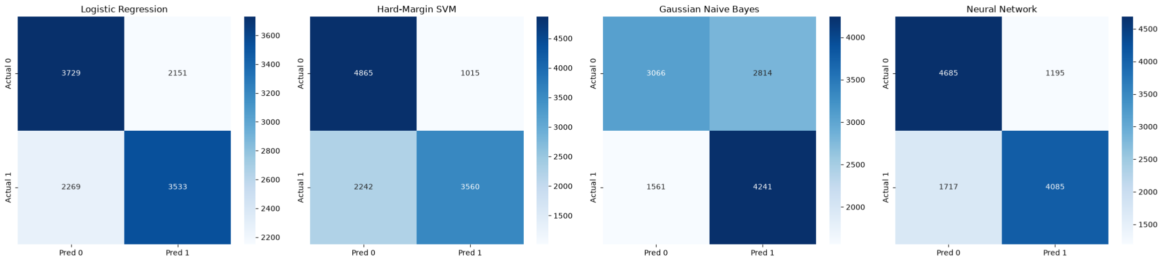

```
Converged in 5000 iterations
```

2. ROC-AUC Curves

```
366/366 ━━━━━━━━━━━━━━━━━━━━ 0s 169us/step
```



3. Confusion Matrices

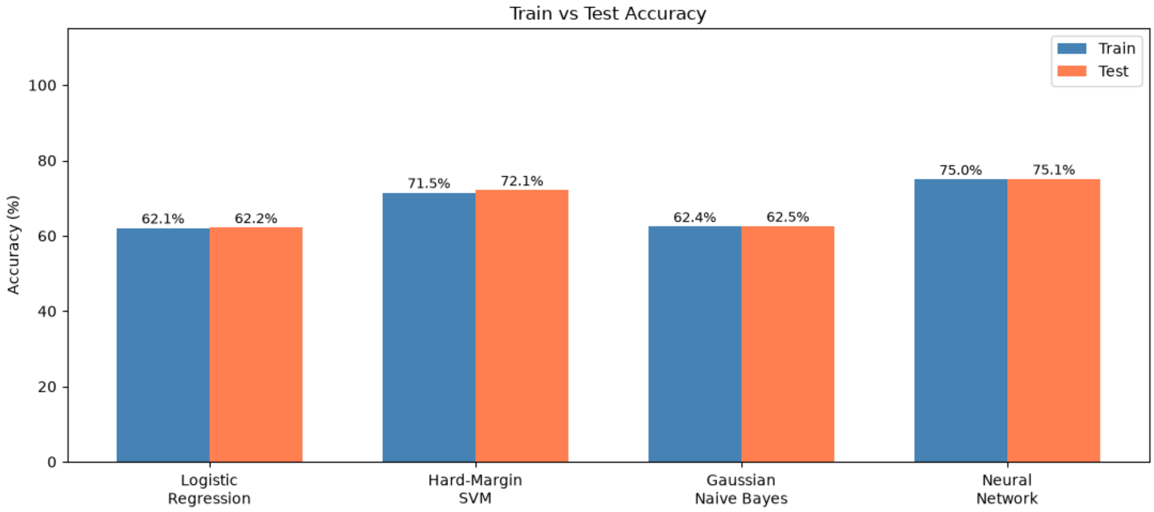


4. K-Fold Cross-Validation — Logistic Regression

```
  0%|                                                                    | 0/500
[00:00<?, ?it/s]/var/folders/hg/9_9zzq252fsdzz4by4y0lvrw0000gn/T/
ipykernel_4793/2760970217.py:30: RuntimeWarning: overflow encountered in
exp
  return 1 / (1 + np.exp(-z))
100%|██████████████████████████████████████████████████| 500/500
[00:00<00:00, 1668.78it/s]
Fold 1: Accuracy=52.64%  F1=57.60%
  0%|                                                                    | 0/500
[00:00<?, ?it/s]/var/folders/hg/9_9zzq252fsdzz4by4y0lvrw0000gn/T/
ipykernel_4793/2760970217.py:33: RuntimeWarning: overflow encountered in
exp
  neg = np.log(np.ones(X.shape[0]) + np.exp(X.dot(self.theta))) -
X.dot(self.theta).dot(y)
100%|██████████████████████████████████████████████████| 500/500
[00:00<00:00, 1591.86it/s]
Fold 2: Accuracy=63.49%  F1=64.88%
100%|██████████████████████████████████████████████████| 500/500
[00:00<00:00, 1572.50it/s]
Fold 3: Accuracy=52.92%  F1=58.96%
100%|██████████████████████████████████████████████████| 500/500
[00:00<00:00, 1668.88it/s]
Fold 4: Accuracy=62.51%  F1=57.07%
100%|██████████████████████████████████████████████████| 500/500
[00:00<00:00, 1631.60it/s]
Fold 5: Accuracy=62.17%  F1=61.50%

K-Fold CV (k=5) Summary — Logistic Regression
Mean Accuracy : 58.75% ± 4.89%
Mean F1-Score : 60.00%  ± 2.88%
```

5. Train vs Test Accuracy

```
Gaussian Naive Bayes Progress: 100%|███████████████| 35046/35046
[00:38<00:00, 901.35it/s]
 315/1096 ━━━━━━━━━━━━━━━━━━━━ 0s 159us/step
1096/1096 ━━━━━━━━━━━━━━━━━━━━ 0s 161us/step
```



Train vs Test Accuracy
Train
Test
Accuracy (%)
0
20
40
60
80
100
62.1%
62.2%
71.5%
72.1%
62.4%
62.5%
75.0%
75.1%
Logistic Regression
Hard-Margin SVM
Gaussian Naive Bayes
Neural Network